

Answer-Type-Aware Evaluation of LLM Hallucinations in Financial Question Answering*

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Abstract

The issue of hallucinations has been recognized as fundamental deficiency in large language models (LLMs). This problem if left unchecked is a huge problem, especially in fields like finance, medicine or law. In this paper, we present a systematic evaluation of multiple LLMs performance on financial question datasets by explicitly separating numerical extraction and reasoning and textual explanation tasks. We evaluate multiple open source LLMs with answer type specific metrics which includes exact-match accuracy and claim-level hallucination rate for all kinds of questions, and numerical and textual questions individually. The LLMs are provided with the context pertaining to the question along with the question and instructed to strictly use only this information. Our results reveal pervasive arithmetic errors in numeric answers and some unsupported claims made in textual explanations. Moreover, no single model performs robustly across both answer types, indicating inherent specialization. These results highlight the limitations of general-purpose LLMs in financial domains and underscore the need for task-aware evaluation and modular model selection in financial QA systems.

1 Introduction

Large Language Models (LLMs) have achieved huge success across a wide range of natural language processing tasks, which lead to their increasing adoption in high-stakes domains like finance. Recent work has explored the use of LLMs for financial queries, report analysis and decision support, where the model must reason over long narratives, huge structured tables and numerical disclosures. Hallucinations or reasoning errors would pose a significant risk in the financial context, like incorrect numerical values being extracted or unsupported textual claims being made, which will directly affect the financial interpretation and downstream decisions.

Prior research has extensively documented the hallucination phenomenon in LLMs. They have proposed taxonomies, detection strategies and mitigation techniques across general application-oriented

*Code and evaluation artifacts are publicly available at <https://github.com/sragvideadalready/financial-llm-hallucinations>

settings[2, 4]. In the financial domain, studies have demonstrated how LLMs quite frequently generate hallucinated or outdated information while answering finance-related questions. [3, 8], while some focused on hallucinations which arise from tabular reasoning and structured financial data [10]. Additionally, recent benchmarks have highlighted the challenges of jointly reasoning over text and tables in retrieval-augmented generation systems [9].

Despite these advances, existing evaluation techniques are largely treating financial question as a unified task, without distinguishing between fundamentally different answer types. In practice, however, financial QA spans over numeric calculation based answers, which require precise arithmetic and multi step reasoning and textual answers, which require faithful grounding in the context and proper factual consistency. Verifying these answer types using the same methods could risk masking the systematic failures like the arithmetic errors which on the surface could seem plausible or the textual hallucinations which are only partially grounded and yet misleading.

Contributions.

- **Answer-Type-Aware Evaluation Framework :** We introduce an evaluation framework for financial question answering which explicitly distinguishes between *numeric* and *textual* answer types. This helps in highlighting the differences in their failure modes.
- **Tailored Verification Methods for Numeric and Textual Reliability :** We implement answer-type-specific verification mechanisms to determine whether a claim is grounded. Numeric answers are verified using T²-RAG benchmark, while textual answers are verified using exact match, token overlap and natural language inference with the document context.
- **Implications for Reliable Financial QA Systems :** Our analysis shows that LLM reliability varies for each model and with each answer-type, which motivates task-aware evaluation protocols for financial question answering.

2 Related Work

Hallucinations in Large Language Models Hallucinations in LLMs have been studied widely, with previous work proposing comprehensive taxonomies, causes and mitigation methods. Surveys by Islam et al. and Li et al. systematically categorize hallucinations into intrinsic and extrinsic types and review mitigation approaches which include retrieval augmented generation, reasoning based methods and agentic systems. These studies explain that hallucinations occur not only from missing information but also from flawed reasoning and overconfident answer generation, even when the relevant context documents are provided. While these papers provide us with a broad conceptual framework they don't focus on domain specific settings like finance.

LLM Hallucinations in Financial Domains Several studies have examined hallucinations specifically in financial applications. Kang and Liu empirically demonstrate that LLMs frequently generate fabricated financial information, highlighting deficiencies in factual consistency and numerical accuracy in finance-related tasks. Shah et al. further show that LLMs often fail to retrieve or reason over financial knowledge beyond reported cutoff dates, leading to outdated or incorrect responses. Complementary work by Zhang et al. introduces FAITH, a framework for assessing hallucinations in financial tables, emphasizing errors that arise when models generate unsupported tabular values.

3 Dataset

3.1 Data Source

We base our evaluation on the FinQA dataset, a widely used benchmark for financial questions answer dataset that contains over 10,000 finance related questions derived from real world financial reports and documents. FinQA primarily focuses on numerical reasoning, with questions grounded in financial tables, textual disclosures and context passages all provided in the dataset.

While FinQA provides high-quality numeric questions and gold numerical answers, it does not include textual answers, nor does it explicitly label answer types, which are crucial for the work done here. These limitations motivate our manual annotation process, described below.

3.2 Dataset Structure

Each example in the dataset is stored in a JSON format and they consist of:

- a natural language question;
- a context passage which includes financial tables, textual descriptions and expectations from relevant financial documents.
- a ground truth answer, which in the original FinQA dataset is numerical

For this study, we extend the dataset by adding:

- a manually annotated answer type label (numeric or textual)
- a human written textual answer for questions where the answer is expressed as a grounded factual explanation.

3.3 Model Visibility Assumptions

To ensure a realistic evaluation environment, models are provided only with:

- the question
- the associated context passage;

Models are not provided with:

- the ground truth answer
- any explanation or reasoning trace
- answer type labels or metadata

This setup will replicate a practical deployment, where models must infer both the answer and the grounding solely from the context given in prompt.

3.4 Manual Annotation Procedure

Since FinQA doesn’t contain textual answers, we manually annotate a subset of questions by carefully reading the associated context passages and tables and building textual answers. In addition each question is manually labeled with its answer type (numeric or textual)

After generating LLM model outputs, we further perform manual annotation of the true labels, where we classify each response as either grounded or hallucinated based on whether the model answer is fully supported by the provided context.

To ensure consistency, all annotations are performed using the same verification criteria applied during evaluation, including context grounding and arithmetic validity wherever applicable.

3.5 Dataset Size and Selection

Rather than evaluating on the full FinQA dataset, we deliberately restrict our study to a smaller high quality dataset of approximately 80 questions. This design choice allows for careful manual annotation and rigorous verification of model outputs, which would be impractical at higher scales.

Table 1: Dataset statistics

Answer Type	Count	Percentage
Numeric	47	58.02%
Textual	34	41.97%
Total	81	100%

4 Methodology

4.1 Models Evaluated

We evaluate open source large language models with differing strengths in general language understanding and mathematical reasoning. Our primary experiments conducted using Qwen2.5-7B[6] , Qwen2-math:7B,[7] and DeepSeek. [5]All models are evaluated independently using the same dataset, prompts and evaluation pipeline to ensure comparability.

4.2 Inference Pipeline

For each model, the full evaluation dataset is processed from end-to-end through the LLM to generate the predicted answers. Each model is provided only with the question and the corresponding formatted context document and tables. The LLM is not given any access to the ground truth answers, or the metadata.

Model outputs are structured and stored in a JSON file of the form `model_answers_modelname.json`, which contains

4.3 Human Annotation of Grounding

Following model inference, all the generated answers are manually annotated to determine whether they are grounded or hallucinated with respect to the relevant context data. An answer is considered grounded

if it is fully supported by the context passage and hallucinated if it is not. These annotations will serve as the true labels for downstream evaluation and are stored in a separate CSV file.

Manual annotation is necessary to capture nuanced hallucination behavior that cannot be reliably detected through automated string-matching alone, especially for the particularly grounded or superficially plausible responses.

4.4 Model Verification Module

To assess the accurateness of the model outputs in a principled manner, we implement an answer type personalized verification module. While the same evaluation metrics are applied across both answer types, verification mechanisms differ for numeric and textual answers.

4.4.1 Numeric Answer Verification

For numeric answers we adopt a verification strategy inspired by the $T^2-RAGbenchmark.priorverification, gold$

Given a predicted numeric answer $pred$ and the gold answer $gold$, we compute:

$$ratio = \frac{pred}{gold}$$

$$scale_adjusted = ratio \times 10^{-round(\log_{10}(ratio))}$$

This scale adjusted ratio enables robust comparison under magnitude shifts commonly observed in financial data. We then check if the scale adjusted ratio is greater than a pre-decided threshold. Numeric predictions that satisfy the verification criteria are labeled as grounded; otherwise they are labeled as hallucinated.

4.4.2 Textual Answer Verification

Textual answers are verified using a combination of checking mechanisms. Specifically, we apply:

1. **Exact match**, to detect verbatim agreement with gold answer.
2. **Token Overlap**, to capture partial lexical alignment.
3. **Natural Language Inference (NLI)**, using RoBERTa-MNLI [1] model to assess whether the generated answer is *entailed*, *neutral*, or contradicted by the context, along with a confidence score.

A textual answer is labeled as grounded if it is either an exact match or classified as *entailed* by the NLI model. Answers classified as *neutral* or *contradicted* are labeled as hallucinated.

5 Evaluation Metrics

5.1 Evaluation Objective

The primary objective of our evaluation is not to assess whether a model’s output is factually correct in an absolute sense. It is to evaluate the faithfulness to the provided input context in the prompt. Specifically, we measure whether the claims produced by the model are grounded and supported by the given context

passage. This framing aligns with real world deployment settings, where we expected models to strictly pertain to the information provided.

5.2 Claim Grounding Accuracy

Claim grounding accuracy will determine the agreement between the predicted grounding relationship and the true grounding for a given claim-context pair.

Let $\hat{t}(x, c)$ denote the predicted relationship between a model-generated claim x and its context c , and let $t(x, c)$ denote the true relationship as determined by manual annotation. Claim Grounding Accuracy is defined as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{t}(x_i, c_i) = t(x_i, c_i)]$$

where N is the total number of evaluated claim-context pairs and $\mathbf{1}[\cdot]$ is the indicator function.

This metric will show the overall correctness of the hallucination classification pipeline. However accuracy alone does not provide us with a full picture of the model behavior, particularly in the presence of asymmetric error costs. therefore, we complement this metric with hallucination rate and error type analysis.

5.3 Hallucination Rate

Hallucination rate measures how frequently a model generates answers that are not grounded in the provided context. This is computed as:

$$\text{Hallucination Rate} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[t(x_i, c_i) = \text{hallucinated}]$$

This metric reflects the tendency of a model to produce unsupported claims. In financial decision-support settings, hallucination rate is a very critical indicator, as even infrequent hallucinations can lead to significant negative consequences when models are used for analysis, or decision-making.

5.4 Error Type Analysis

To better understand model behavior we employ one more metric to perform an error type analysis by categorizing misclassifications into following groups:

- **False Positives (FP):** Claims that are grounded in the context but are incorrectly classified as hallucinated.
- **False Negatives (FN):** Claims that are not grounded by the context but are incorrectly classified as grounded.

Among these two, false negatives are more harmful, as they correspond to unsupported claims being presented as legitimate claims. This failure mode is concerning in high-risk domains like finance, where unsupported claims will directly influence decision making.

Reproducibility. All code used for model inference, answer verification, and metric computation is available at <https://github.com/sragvideadalready/financial-llm-hallucinations>.

6 Results

We evaluate three open source LLMs- **Qwen2.5-7B**, **Qwen2-Math-7B**, and **DeepSeek-R1** on a manually annotated subset of 81 financial question- context pairs. we report overall metrics as well as answer type specifics results for numeric and textual answers.

6.1 Overall Performance

Evaluation across all answer types reveals notable differences in grounding across models.

- **Qwen2.5-7B** achieves an overall grounding accuracy of 89% and a hallucination rate of 63%, producing 3 false positives and 6 false negatives.
- **Qwen2-Math-7B** attains the lowest overall accuracy of 84% and the highest hallucination rate at 74% with 11 false positives and 2 false negatives.
- **Deepseek-R1** matches Qwen2.5-7B in overall accuracy 89%, while exhibiting lowest hallucination rate of 50.6%, with 5 false positives and 4 false negatives.

Although aggregate accuracy values appear relatively high, hallucination rate remains substantial across all models, indicating that correct grounding cannot be inferred from accuracy alone.

Table 2: Overall performance across models

Model	Accuracy	Hallucination Rate	FP / FN
qwen2.5:7b	88.9%	63.0%	3 / 6
qwen2-math:7b	84.0%	70.4%	11 / 2
deepseek-r1	88.9%	50.6%	5 / 4

6.2 Numeric Answers Performance

All three models demonstrate high numeric accuracy but extremely poor numeric grounding reliability.

- **Qwen2.5-7B** achieves a numeric accuracy of 93.6% , yet exhibits a numeric hallucination rate of 89.4%.
- **Qwen2-Math-7B** records a numeric accuracy of 91.5% but shows the highest numeric hallucination rate at 95.7%
- **DeepSeek-R1** attains a numeric accuracy of 93.6% , with a comparatively lower, though still substantial,numeric hallucination rate of 68.1%.

These results indicate that models frequently produce numerically formatted outputs that appear correct, while failing verification against grounded financial context. Percentage and ratio based questions are particularly error-prone, with models often generating outputs that differ from gold values by several orders of magnitude.

Table 3: Numeric answer grounding performance

Model	Accuracy	Hallucination Rate
qwen2.5:7b	93.6%	89.4%
qwen2-math:7b	91.5%	95.7%
deepseek-r1	93.6%	68.1%

6.3 Textual Answers Performance

Textual answers exhibit lower accuracy but significantly better grounding behavior compared to numeric answers.

- **Qwen2.5-7B** achieves a textual accuracy of 82.4% and a hallucination rate of 26.5% .
- **Qwen2-Math-7B** performs worst on textual questions, with an accuracy of 73.5% and a hallucination rate of 35.3%
- **DeepSeek-R1** attains a textual accuracy of 82.4% and a hallucination rate of 26.5%, matching Qwen2.5-7B.

The introduction of a conservative NLI entailment threshold (0.7) reduces spurious grounding classifications, leading to a modest decrease in textual accuracy while improving reliability.

Table 4: Textual answer grounding performance

Model	Accuracy	Hallucination Rate
qwen2.5:7b	82.4%	26.5%
qwen2-math:7b	73.5%	35.3%
deepseek-r1	82.4%	26.5%

6.4 Error Type Analysis

Error analysis reveals asymmetric failure across models.

- **False Negatives**, correspond to hallucinated claims incorrectly classified as grounded, are present in all the models (6 for Qwen2.5-7B, 2 for Qwen2-Math-7B, and 4 for DeepSeek-R1).
- **False Positives** are especially pronounced for Qwen2-Math-7B (11 instances), indicating a conservative grounding classification under textual verification.

False negatives are particularly concerning in financial settings, as they correspond to unsupported claims being presented as legitimate information.

6.5 Summary of Findings

Across all the models evaluated, we observe a constant discrepancy between numeric accuracy and numeric faithfulness. While models frequently generate plausible numeric outputs, verification shows pervasive grounding failures. In contrast textual answers, even though less accurate, exhibit substantially lower hallucination rates under the conservative entailment criteria.

Among the evaluated models, DeepSeek-R1 demonstrates the most balanced and optimal performance, achieving the lowest hallucination rate while maintaining a competing accuracy. However, none of the models evaluated can be considered reliably grounded across both numeric and textual financial question answering tasks.

7 Error Analysis

We dug into how each model handles false positives and false negatives when grounding predictions. Table X tells the story: Qwen2.5:7b hardly ever flags something as grounded when it isn't (just 3 false positives), but it's quick to reject claims that are actually correct—6 times, in fact. It plays things safe, leaning conservative. On the other hand, Qwen2-Math:7b swings the opposite way. It racks up 11 false positives, so it often accepts claims as grounded even when they shouldn't be, though it rarely rejects grounded claims (just 2 false negatives).

DeepSeek-R1 sits somewhere in the middle: 5 false positives, 4 false negatives. Its error pattern matches its lower hallucination rate and solid accuracy. This kind of balance matters in real-world settings, where missing true claims or letting hallucinations slip through can both cause problems.

Looking at the types of answers, numeric errors mostly happen because of hallucinated values, not because the numbers are just a little off. Qwen models, even when they get the answer right, still hallucinate numbers a lot. Textual errors are a bit different—they're spread out more and often come from cases where the answer almost matches, like paraphrasing or partial overlap, but doesn't line up word-for-word with the gold answer.

7.1 Discussion

There's a clear split between how these models handle grounding for numbers versus text. All three get pretty high numeric accuracy, but hallucination is still a big issue—especially for Qwen2.5:7b and Qwen2-Math:7b. So, even if the number looks right, that doesn't mean the model actually found evidence for it. "Good enough" numeric answers can still be hallucinated.

DeepSeek-R1 stands out. It keeps hallucination rates low on both numeric and textual tasks, showing it's not just good at getting the right answer—it's better at sticking to the evidence. Even when DeepSeek-R1 and Qwen2.5:7b score the same on textual accuracy, DeepSeek-R1 hallucinates less. That really drives home that grounding and correctness aren't the same thing.

Qwen2-Math:7b is supposed to be great at math, but it struggles with grounding for text and has the worst hallucination rate overall. Turns out, being good at number-crunching doesn't guarantee the model can actually back up its answers when it comes to sticking to the facts.

The bottom line: we need to check numeric and textual grounding separately, and we can't just look at accuracy. Explicit hallucination metrics matter.

7.2 Limitations

This study isn't without its flaws. For one, the dataset is pretty small—only 81 samples—so while the trends are clear, we need larger tests to really pin down model differences.

Second, we labeled grounding by hand. Even with careful checking, some subjectivity sneaks in. The verification module helps, but we might still miss subtle errors.

Third, for numbers, we use a scale-adjusted ratio to account for magnitude differences. This helps when units change, but it could still call partially correct answers “hallucinated” if the model doesn’t explain the reasoning.

Finally, these models worked offline in a single-pass setup. We didn’t use retrieval, reranking, or multi-step verification. Performance might shift in systems that add those extra layers.

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